

AYA MOURAD

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RESEARCH SUMMARY

PhD researcher and EU Marie Curie Fellow in NLP and large language models. My work targets low-resource languages, where I design compute-efficient, long-context methods for information extraction (nested NER, coreference, relation extraction, event detection) and build the datasets and benchmarks to evaluate them, achieving strong results without heavy compute or large training corpora.

EDUCATION

Sorbonne Université

PhD, École Doctorale Informatique, Télécommunications et Électronique
ISIR – MLIA Team | Marie Skłodowska-Curie COFUND Fellowship (SOUND.AI)

Paris, France

Nov 2024 – Oct 2027

American University of Beirut

Master of Science in Computer Science (GPA 4.0)

Beirut, Lebanon

Feb 2019 – Mar 2022

American University of Beirut

Bachelor of Science in Computer Science

Beirut, Lebanon

Sep 2012 – May 2016

PUBLICATIONS

Forthcoming

A. Mourad, M. Jarrar. “ADABNER: Arabic Digital Archive Books with Nested Entity Recognition.” *Accepted at ACL 2026*. | Dataset: [Zenodo DOI: 10.5281/zenodo.19468385](#)

One additional paper on open-vocabulary topic annotation for multilingual Wikipedia is under review (EMNLP 2026); title withheld during the anonymity period.

Published

A. Mourad, F. K. Abu Salem, S. Elbassuoni. “Detecting gender bias in Arabic text through word embeddings.” *PLoS One*, 20(3), e0319301, 2025.

A. Mustapha, H. Al-Khansa, H. Al-Mubasher, **A. Mourad**, et al. “ARASTEM: A Native Arabic Multiple-Choice Question Benchmark for Evaluating LLMs Knowledge in STEM Subjects.” *arXiv:2501.00559*, 2024.

S. Elbassuoni, H. Ghattas, J. El Ati, et al. (incl. **A. Mourad**). “Capturing children food exposure using wearable cameras and deep learning.” *PLOS Digital Health*, 2(3), e0000211, 2023.

A. Mourad, S. Elbassuoni. “A Large-Scale Analysis of COVID-19 Tweets in the Arab Region.” *Social Network Analysis and Mining*, 12(1), 71, 2022.

PROFESSIONAL EXPERIENCE

Sorbonne Université – ISIR, MLIA Team

PhD Researcher

Paris, France

Nov 2024 – Present

- Developing the Arabic information extraction research pipeline (nested NER, event detection, coreference, relation extraction); currently building a compute-efficient architecture that scales to long documents.
- Built **ADABNER**, the first large-scale nested NER corpus for Modern Standard Arabic literary books (138 books, ~876K tokens, 21 entity types), and benchmarked BERT encoders against few-shot LLMs.
- Funded by the highly competitive EU Marie Skłodowska-Curie COFUND Fellowship (SOUND.AI).

Independent Research Collaboration

NLP Researcher (Under review, EMNLP 2026)

Remote

Apr 2025 – May 2026

Collaborators: Christophe Khalil (AUB), Dr. Shady Elbassuoni (AUB)

- Designed a decoupled GPT-4.1 / Claude annotation pipeline, annotating 46,908 multilingual Wikipedia lead sections with 31,799 open-vocabulary topics.
- Validated judge scores via embeddings, human audit ($\kappa=0.94\text{--}0.98$), and Wikidata grounding.
- Showed zero cross-lingual bias amplification ($\Delta\text{AUC} \approx 0$) via one-vs-rest and MMD probes.

Updated: June 2026

American University of Beirut – AI Hub

Research Assistant

Beirut, Lebanon

Apr 2024 – Nov 2025

- Developed **AraSTEM**, a native Arabic multiple-choice benchmark to evaluate LLM knowledge in STEM subjects; led dataset curation, question formulation, and model benchmarking.
- Built a Retrieval-Augmented Generation (RAG) system to enhance LLM knowledge in Arabic medical contexts.
- Organized the 2024 Summer AI Academy (AI fundamentals and ML for high-school students) and developed four modules for a graduate Large Language Models course (Spring 2025): embeddings, tokenization, prompt engineering, and AI ethics.

American University of Beirut – Computer Science Department

Research & Teaching Assistant

Beirut, Lebanon

Mar 2022 – Apr 2024

- Developed a deep learning approach for food-item detection and healthiness prediction (MobileNetV2, InceptionV3, YOLOv3), contributing to work published in *PLOS Digital Health* (2023).
- Conducted research on gender bias detection in Arabic word embeddings (WEAT, Direct Bias); published in *PLoS One* (2025).
- Investigated hate speech across social media geolocated in Lebanon using GPT-4, CNN, RNN, and BERT.
- Assisted with the Professional Diploma in AI & Data Science (Machine Learning, Data Science, Deep Learning, Introduction to Programming).

American University of Beirut

Graduate Research Assistant

Beirut, Lebanon

Feb 2021 – Dec 2021

Thesis: A Large-Scale Analysis of COVID-19 Tweets in the Arab Region

Graduate Teaching Assistant

Nov 2018 – Dec 2021

- Assisted with Advanced Machine Learning, Artificial Intelligence, Web Programming & Design, Computer Networks, and Introduction to Programming.

American University of Beirut

Developer and Data Analyst

Beirut, Lebanon

Feb 2017 – Aug 2020

- Built faculty software for the Faculty of Health Sciences (performance dashboards, KPI monitoring across 3 cycles / 15 years, admissions and enrollment tracking) with SQL Reporting Services (SSRS) and QlikView, and led migration to a paperless Laserfiche/OCR repository.

FELLOWSHIPS & TRAINING

- **Marie Skłodowska-Curie COFUND Fellow (SOUND.AI)**: European Commission, Sorbonne Université, 2024–2027. Highly competitive EU-funded interdisciplinary AI research programme. Awarded competitively after defending an original PhD research proposal before a 15-member jury.
- **Deep Learning School 2025**: EFELIA Côte d’Azur, Sophia Antipolis, France, Jun–Jul 2025.
- **Google Cloud Research Credits**: research compute award supporting Vertex AI experiments.
- **AUB Graduate Assistant Fellowship**: Merit-based award covering 100% of tuition.

TECHNICAL SKILLS

Programming Languages & Scripting: Python, R, C#, SQL, PHP, L^AT_EX

AI & ML frameworks: PyTorch, TensorFlow, HuggingFace Transformers, PEFT, vLLM, LangChain, OpenCV

LLMs & Methods: Fine-tuning, RAG, few-shot in-context learning and prompt engineering, multi-GPU / distributed training, LLM evaluation & benchmarking, dataset & annotation design

Infrastructure: Vertex AI, Google Cloud (GCP), HPC / GPU clusters, Linux, Git, Jupyter

Research Areas: Information Extraction (nested NER, coreference, relation extraction, event detection, entity linking), Multilingual NLP & Low-Resource NLP, AI Agents, Recommender Systems, Information retrieval

Languages: Arabic (Native), English (Full Professional)